Arguing with Multiple Semiotic Modes

Pathways to Multimodality

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Volume 2

Arguing with Multiple Semiotic Modes

Theoretical Issues and Empirical Advances in Multimodal Argumentation Studies

Edited by
Hartmut Stöckl, Assimakis Tseronis and Janina Wildfeuer

DE GRUYTER

Open access publication supported by the Open Access Book Fund of the University of Groningen (NL), the University of Salzburg (AT) Publication Fund, and Örebro University (SE).

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ISSN 2941-380X
ISBN 978-3-11-914424-7
ISBN 978-3-11-222117-4 (PDF)
ISBN 978-3-11-222129-7 (E-PUB)
DOI https://doi.org/10.1515/9783112221174

Library of Congress Control Number: 2026936248

Bibliographic information published by the Deutsche Nationalbibliothek
The Deutsche Nationalbibliothek lists this publication in the Deutsche Nationalbibliografie; detailed bibliographic data are available on the Internet at https://dnb.dnb.de.

© 2026 the author(s)/editor(s), published by Walter de Gruyter GmbH, Genthiner Straße 13, 10785 Berlin, Germany

De Gruyter and Walter de Gruyter GmbH are part of De Gruyter Brill.
www.degruyterbrill.com

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Printing: CPI books GmbH, Leck

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Jean H. M. Wagemans

The Multimodal Argument Type Identification Procedure (MATIP)

A parametric approach to identifying multimodal argument types

Abstract: The chapter introduces a novel method for identifying argument types (schemes) in multimodal arguments, called the 'Multimodal Argument Type Identification Procedure' (MATIP), based on the argument categorization framework of the 'Periodic Table of Arguments' (PTA) and inspired by the 'Argument Type Identification Procedure' (ATIP) that was originally developed for verbal arguments. By facilitating cross modal aggregation and analysis of argumentative elements, the procedure avoids the need for translating non-verbal information into its verbal counterpart(s) and mitigates the complexities of reconstructing arguments based on matching with predefined argument schemes. The author elucidates the theoretical and methodological starting points of the MATIP and exemplifies its functionality and outcomes through the examination of a sample dual code argument integrating verbal and visual components.

Keywords: argument scheme, argument type, crossmodal aggregation, multimodal argumentation, Multimodal Argument Type Identification Procedure (MATIP), Periodic Table of Arguments (PTA)

1 Introduction

How can arguments that include non-verbal components, such as visuals, be conceptualized in terms of argument types (schemes)? In this chapter, I propose an answer to this methodological question by introducing the Multimodal Argument Type Identification Procedure (MATIP), an innovative method for identifying multimodal argument types based on the argument categorization framework of the Periodic Table of Arguments (PTA) (Wagemans, 2016, 2023) and its associated Argument Type Identification Procedure (ATIP) (Wagemans, 2025).

In the field of argumentation theory, there is broad consensus on the methodological starting point that individual arguments can be evaluated by means of asking so-called 'critical questions' (see, e.g., Walton et al., 2008, pp. 15–17; van Eemeren et al., 2014, pp. 19–21). As these questions differ per argument type, the evaluation must be preceded by an identification of the type of argument under

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scrutiny in terms of a taxonomy of arguments. Several such taxonomies have been proposed, usually consisting of a set of predefined argument schemes, each with a conclusion and a set of premises, all expressed in the verbal mode. However, using these taxonomies to identify the type of a given multimodal argument necessitates the analyst to presuppose that (1) the multimodal material contains the same argumentative elements as those described in the scheme and (2) the meaning of each non-verbal element is equivalent to that of its verbal counterpart. Both presuppositions are problematic, and in this chapter, I indicate how the analyst can avoid making them by using the argument categorization framework of the PTA and its associated MATIP instead.

The challenge of identifying multimodal arguments is situated within the broader academic debate on multimodal argumentation that has been ongoing for three decades – see, e.g., Birdsell and Groarke (1996, 2007), Kjeldsen (2015), Groarke, Palczewski, and Godden (2016), Tseronis and Forceville (2017), Gonçalves-Segundo, Macagno, and Azevedo (2021), and Stöckl (2024). Among the central issues in this debate are the following questions: (1) Do multimodal arguments exist and, if so, in what way are they multimodal? (2) To what extent can or should they be represented in verbal terms? (3) How should their argumentative quality be evaluated? To contextualize my proposal for using the MATIP to identify multimodal argument types, I will briefly outline my position on these three fundamental issues.

Regarding the first issue, some authors state that arguments can only be expressed through discursive means and, therefore, deny the existence of non-verbal argumentation (see, e.g., Gonçalves-Segundo, Macagno, & Azevedo, 2021, pp. 723–725). The exact phrasing of their positions is of some importance here. Fleming (1996) and later Patterson (2010), for example, are of the opinion that *purely* visual arguments do not exist. They acknowledge that non-verbal elements may convey aspects of arguments but contend that these elements must be reconstructed in verbal terms to clarify their argumentative function, thereby losing their distinctly non-verbal features. However, as no authors define visual arguments in this way (Groarke, Palczewski, & Godden, 2016, p. 218; Groarke, 2019, pp. 335–336), these critics seem to be attacking an empty castle. Groarke notes that, in practice, argumentative exchanges often display a combination of verbal and non-verbal elements: 'In real life arguing, arguers typically mix words and visuals and whatever other modes of expression (non-verbal sounds, music, etc.) which can be used to make a compelling case for a conclusion they propose' (2019, p. 335). I hold this observation to be correct and will take it as a starting point for this chapter that arguments that include both verbal and non-verbal components can be classified as *multimodal* arguments.

This brings us to the second issue. If arguments can be expressed, wholly or partially, in a non-verbal manner, to what extent can or should they be translated into verbal terms? Is such a translation even possible? And if it is, do we risk losing relevant information by representing visuals as propositions or by making a choice between the various propositions made available, for instance, by an image (see, e.g., Kress & van Leeuwen, 1996). These questions are discussed under the heading of the so-called 'propositionality issue' (Kjeldsen, 2015, p. 119). In my view, this issue originates from the fact that argumentation theorists have already developed quite an extensive set of methods for analyzing and evaluating arguments expressed in the verbal mode. Believing it would be a missed opportunity to leave the existing armamentarium unused in studying multimodal argumentation, these theorists set out to design a method for translating non-verbal elements into verbal forms.

While I acknowledge that a lossless translation of the non-verbal components of a multimodal argument into the verbal mode is impossible, I am also convinced that the method presented in this chapter renders the need for such a translation obsolete. Since the MATIP is based on the idea that an argument consists of a specific combination of references and predications, each of which can be expressed in any modality, it operates at a level of aggregation that incorporates both verbal and non-verbal components. By facilitating this 'crossmodal aggregation' of the components of an argument, the MATIP does not require the analyst to translate non-verbal components into verbal ones.

The third issue concerns how to assess the acceptability of multimodal arguments. As noted earlier, there is general consensus that evaluating an individual argument must be preceded by identifying its type in terms of a given taxonomy of arguments. The main reason for following this sequence is that such an evaluation requires the formulation of criteria that are partially type-dependent (e.g., an argument from cause is assessed by means of other criteria than an argument from authority). In this chapter, I will not provide these specific assessment criteria directly but instead indicate how the evaluative questions addressing the 'points of attack' of a given multimodal argument can be systematically generated based on an identification of its type.

The chapter is structured as follows. In Section 2, I review taxonomies of arguments that have been proposed as suitable for analyzing multimodal argumentation. I highlight two main issues arising from applying these taxonomies to identify multimodal argument types, which I label as the 'matching' problem and the 'equivalence' problem. Then, in Section 3, I outline the theoretical and methodological starting points of the Periodic Table of Arguments (PTA) and its associated Argument Type Identification Procedure (ATIP). In Section 4, I present the MATIP,

detailing the steps of the procedure and demonstrating its functionality and outcomes through an example analysis of a multimodal argument that combines verbal and visual elements. I also explain a method for generating critical questions that can be used to evaluate multimodal arguments after their type has been identified. Finally, in Section 5, I reflect on the extent to which the MATIP alleviates the problems associated with existing methods for multimodal argument type identification and propose directions for further research.

2 Problems of existing taxonomies

In the field of argumentation theory, several taxonomies of argument schemes have been developed – for an overview, see Wagemans (2022, pp. 584–586). Regarding their use in analyzing and evaluating multimodal argumentation, Gonçalves-Segundo, Macagno, and Azevedo (2021), as well as Stöckl (2024), discuss a selection that includes the new dialectical taxonomy proposed by Walton, Reed, and Macagno (2008), the pragma-dialectical taxonomy developed by van Eemeren and Grootendorst (2004), and the Argumentum Model of Topics (AMT) introduced by Rigotti and Greco (2019). In what follows, I will briefly examine these three taxonomies, focusing on their suitability for identifying multimodal arguments.

The new dialectical taxonomy proposed by Walton, Reed, and Macagno (2008) is empirically based, providing ad hoc descriptions of naturally occurring schemes composed of a conclusion and a varying number of premises (Wagemans, 2016, pp. 2–3). It has been criticized for lacking a theoretical rationale to distinguish between argument schemes as well as for its unsystematic approach to defining both the constitutive elements of the schemes and their associated critical questions. The critical questions address various aspects of the schemes but do so inconsistently: while some schemes include premises that are not addressed in the evaluation, others feature critical questions that are unrelated to any premises but focus on the context in which the argumentation occurs (Hornikx, 2013, pp. 134–135; de Jong, 2019). Furthermore, the critical questions require varying types of answers – sometimes a simple yes or no, sometimes gradual responses, and in other cases more elaborate explanations (Wagemans, 2011).

The pragma-dialectical taxonomy developed by van Eemeren and Grootendorst (2004) consists of three main types and a varying number of subtypes. While this taxonomy is more systematic than the new dialectical taxonomy, it too suffers from several problems: the subtypes are not always accurate specifications of the main types, there are inconsistencies in the formulations of the bridging (or miss-

ing) premise, and some critical questions are irrelevant or underspecified and therefore inadequate for evaluating arguments of a particular type or subtype (Hitchcock & Wagemans, 2011).

The Argumentum Model of Topics (AMT) (Rigotti & Greco, 2019) describes a large number of argument structures, each of which consists of six interrelated elements. The AMT is more systematic than the new dialectical taxonomy and more detailed than the pragmadialectical one. Its practical application, however, requires laborious and complex reconstruction efforts that only expert analysts can perform. Since it is rare for all six elements to be explicitly present in the discourse, the analyst must supplement the structure using contextual information and background knowledge, thereby overcommitting the analyst and inviting them to engage in the risky business of ‚*hineininterpetieren*' (i.e., over-interpreting).

Apart from the specific issues of the above taxonomies, their use for identifying argument schemes faces two more general challenges. First, there is the difficulty of matching all the argumentative elements described in the scheme with those present in the discourse. The taxonomies typically consist of a more or less organized list of argument schemes, each characterized by a set of idealized properties. To identify the type of a given argument, the analyst compares the descriptions of argument types in the list with the actual argument as expressed in the discourse and determines the best match based on observed similarities.

One major drawback of this comparative approach is the variation in identification quality that comes with the fact that some natural arguments may align better with the idealized descriptions than others. If there is no perfect match, the resulting identification may only provide an imprecise or incomplete description of the argument. What are the exact characteristics of an argument that 'largely' resembles a predefined argument scheme? And if there is no complete match, does that mean the analyst found a new argument scheme?

Another flaw is the reliance on subjective interpretations of similarities between the argument types in the taxonomy and the argument in the discourse. When the idealized descriptions extend to contextual aspects of the discourse, proper identification may even be impossible due to insufficient information. Since these consequences result from the chosen identification method, in which the analyst is trying to match idealized descriptions with elements contained in the discourse, I will refer to them as consequences of the 'matching' problem – see also Wagemans (2020, pp. 2–3) and Hinton and Wagemans (2022, p. 62).

Furthermore, applying the aforementioned taxonomies to identify the type of argument encounters what I will term the 'equivalence' problem. This problem arises when the analyst attempts to recognize the verbally expressed elements of the argument type included in the taxonomy within the multimodal material. In as

far as these elements are expressed in a non-verbal mode, the analyst must find a corresponding verbal equivalent, thereby assuming that the meaning of the non-verbal elements is identical to the meaning of their verbal counterparts. This assumption is problematic because translating non-verbal elements may transform or reduce their meaning, influencing the subsequent evaluation of the argument. This process allows for subjective variation in identifying the points of attack – the aspects of argumentation under evaluation – and may lead to disagreements about which criteria are relevant for assessing these points. By mentioning these consequences, I do not mean to criticize the very possibility of disagreement among evaluators about the quality of an argument as such. This possibility is always present, and unproblematic. Rather, I would like to emphasize that in any *systematic* evaluation procedure, disagreements should not pertain to the objects of evaluation nor to the criteria used to assess their quality, but only to the actual outcome of the evaluation resulting from applying these criteria.

In summary, the use of these existing taxonomies to identify argument types is challenged by the 'matching' problem and the 'equivalence' problem. The matching problem involves imprecise or subjective identifications when comparing idealized descriptions to real discourse. The equivalence problem arises from translating non-verbal elements into verbal counterparts, risking meaning reduction and inconsistencies in evaluation. These issues reveal the limitations of these taxonomies, introducing subjectivity and disagreement not only in the identification process itself but also in defining the elements and criteria on the basis of which they are evaluated.

3 The PTA and the ATIP

The argument categorization framework of the Periodic Table of Arguments (PTA) integrates the traditional dialectical accounts of argument schemes and fallacies as well as the rhetorical accounts of logical, ethotic, and pathetic means of persuasion into a systematic and comprehensive whole (Wagemans, 2016, 2023). Like other such taxonomies, the PTA can be applied for analyzing, evaluating, and producing argumentative discourse. Both its method for categorizing arguments and for identifying the type of a given argument, however, radically differ from the ones used in the other taxonomies. For the present purposes, I will highlight the two most salient differences.

First of all, the PTA-based method does not require the analyst to look for argument schemes with some predefined set of properties and match those with the argument under scrutiny, but to follow a step-by-step procedure that reveals the

characteristics of that argument on the basis of the actual material presented in the discourse. This so-called Argument Type Identification Procedure (ATIP) (Wagemans, 2025) starts from the explicit elements of the discourse and derives, if needed, implicit elements from them. The application of the procedure results in an identification of the type of argument in terms of the PTA.

Second, the categorization framework of the PTA and its associated ATIP take a parametric approach to argument type identification. Commonalities and differences between argument types are reflected as commonalities and differences in the values of specific parameters that express basic characteristics. In its current version, the PTA works with three such parameters: (1) the argument form, which is the configuration of references (subjects) and predications (predicates) in the statements expressing the conclusion and premise of the argument; (2) the argument substance, which characterizes the semantic content of these statements; and (3) the argument lever, which expresses the underlying mechanism of the argument.

The first parameter, the argument form, can take four different values, which are indicated with Greek letters alpha, beta, gamma, and delta. The specific value of this parameter can be obtained by determining the reference and the predication of the statements expressed in the conclusion and the premise of the argument under scrutiny. In Table 1, for each of these values, I listed the corresponding configuration of references (a, b, …) and predications (X, Y, …) as well as an annotated example. The ATIP describes in more detail how an analyst can determine the argument form based on a standardization of the statements expressed in the conclusion and premise and an identification of their references and predications (Wagemans, 2025, pp. 4–8).

Tab. 1: The values of the parameter 'argument form'

configuration of references and predications	value of argument form	annotated example
a is X because a is Y	alpha	She (a) must love Patricia (X), because she (a) is looking at her all the time (Y)
a is X because b is X	beta	Members of the Dutch Royal House (a) should pay taxes (X), because every ordinary citizen in the Netherlands (b) pays taxes (X).
a is X because b is Y	gamma	Peace (a) can be started with truth (X) because wars (b) can be started with lies (Y).
q [is A] because q is Z	delta	The president did a great job (q) [is acceptable (A)] because this (q) was stated by *The Economist* (Z).

The second parameter is the argument substance, which reflects the semantic content of the statements involved. The set of values this parameter can take depends on the argument form (see Table 2): for alpha form arguments, it reflects the predicate types employed in the conclusion and the premise; for beta arguments, the subject types; for gamma arguments, the relationships between the subjects and between the predicates; and for delta arguments, the point of view employed in the premise. The ATIP describes in more detail how the analyst can determine the value of the argument substance by using definitions and examples (Wagemans, 2025, pp. 9–11).

Tab. 2: Values of the parameter 'argument substance' depending on the form

form	determinant of substance	values of substance
alpha	predication type	fact (F), value (V), policy (P)
beta	reference type	singular (s), particular (p), universal (u)
gamma	relations between references and predications	identical (i), different (d)
delta	perspective of the premise	first person (I), second person (II), third person (III)

The argument substance of the first example in Table 1, for instance, is FF, as both the conclusion and the premise are statements of fact.

The third parameter, the argument lever, expresses the underlying mechanism of the argument and plays an important role in the assessment of the quality of the argument (see Hinton & Wagemans, 2022; Hornikx & Wagemans, 2023, pp. 186–188). Its value takes the form of a keyword expressing the relationship between specific elements of the conclusion and premise. As the lever often remains implicit, the ATIP describes how to derive the lever from the explicit material in the discourse in case it has remained implicit (Wagemans, 2025, pp. 12–14). In the first step, the analyst determines the so-called abstract lever, which is based on the argument form. In the second step, the information contained in the argument substance, the context in which the argument is put forward, and our knowledge of the world is combined to formulate the so-called concrete lever. In Table 3, I listed the abstract and concrete levers of the examples given above, indicating the keyword in capitals, and the traditional name of the argument (which is often based on that keyword).

Tab. 3: Values of the parameter 'argument lever' and of the argument names for the four examples

example	value of argument lever including the KEYWORD	argument name
She (a) must love Patricia (X), because she (a) is looking at her all the time (Y)	looking at her all the time (Y) is a SIGN of must love Patricia (X)	argument from sign
Members of the Dutch Royal House (a) should pay taxes (X), because every ordinary citizen in the Netherlands (b) pays taxes (X).	Members of the Dutch Royal House (a) are EQUAL to every ordinary citizen in the Netherlands (b)	argument from equality
Peace (a) can be started with truth (X) because wars (b) can be started with lies (Y).	Peace (a) is the OPPOSITE of wars (b); can be started with truth (X) is the OPPOSITE of can be started with lies (Y)	argument from opposites
The president did a great job (q) [is acceptable (A)] because this (q) was stated by *The Economist* (Z).	was stated by *The Economist* (Z) is EPISTEMICALLY AUTHORITATIVE of [being acceptable (A)]	argument from expert opinion

The parametric approach to argument classification defines an 'argument type' in a purely descriptive way, i.e., as a summary of the values of the parameters reflecting several basic properties of the argument: its form, substance, and lever. Two arguments belong to the same type if they share all these values, and they belong to different types if at least one of these values is different. While the PTA is an overview of various argument types distinguished along these lines, the associated ATIP is a step-by-step procedure for identifying the type of any given argument, conceived of as a combination of two statements, one of which is functioning as the conclusion and the other as the premise of the argument.

4 From ATIP to MATIP

As discussed in the previous section, the theoretical framework of the PTA was originally developed to categorize arguments presented as verbal statements. In this section, I will show how multimodal arguments can be incorporated into the PTA framework and how the ATIP can be modified into a procedure tailored to identifying multimodal argument types by introducing the necessary adaptations. To differentiate this new procedure from the original ATIP, I will refer to it as the Multimodal Argument Type Identification Procedure (MATIP). I will explain its steps and demonstrate its application by using an example of a multimodal argument.

The main reason why the PTA is suitable for multimodal argument categorization is that it works with the notion of 'statement' as an attribution of a predication (predicate) to a reference (subject). Its conceptualization of an argument as a combination of two of such statements, therefore, allows for what I will call the 'crossmodal aggregation' of argumentative elements: the characteristics of an argument that are decisive for identifying its type are analyzed and determined *independent* of the mode in which the statements constituting the argument are expressed.

In verbal statements such as 'the president did a great job', for instance, 'the president' functions as the reference and 'did a great job' as the predication. In the case of multimodal arguments, as Groarke (2019) has shown, nothing prevents the analyst from interpreting a picture as a statement, and from finding relevant connections between the pictorial statement and the verbal statement by determining which references and predications are relevant from an argumentative point of view. I will take Groarke's example of the 'Neuschwanstein argument' to illustrate this point.

> Consider a situation in which my wife suggests that we should go to see Neuschwanstein Castle (the famous Bavarian castle built by King Ludwig II) on a trip to Germany. When I question her suggestion, she tries to convince me that it is something worth seeing by showing me the photograph in Figure 5. In doing so she provides me with a reason for concluding that we should visit Neuschwanstein, though she does so visually – appropriately so given that the issue at question is what we should go to see.



Figure 5. Neuschwanstein Castle at Dusk

> We can describe the argument by saying that my wife has provided me with a visual premise (a visual reason) in support of her contention that "We should go to see Neuschwanstein Castle." (2019, pp. 345-347, original italics)

Groarke reconstructs this multimodal argument by means of a so-called 'Key Components Table (KC Table)' containing the verbal claim 'We should see Neuschwanstein' as the conclusion and the picture as its premise, the latter accompanied by the explanation that it is a 'photograph of Neuschwanstein at dusk' – see Table 4.

Tab. 4: Groarke's (2019, p. 347) KC Table of the Neuschwanstein argument

Key Components	**Role**	**Explanation**
	Premise (**p**)	Visual (photograph of Neuschwanstein at dusk)
"We should visit Neuschwanstein"	Conclusion (**v**)	Verbal claim

As this analysis of the components of a multimodal argument partially reflects the first two steps of MATIP, I will now turn to specifying the steps of this procedure and indicate which results they yield. Also, I will indicate specifications in relation to the ATIP, the latest iteration of which is outlined in Wagemans (2025).

Step 1. Analyze the statements
In this step, the statements are analyzed. This entails determining their function (conclusion or premise) and breaking them down into their elements (reference and predication). From the KC Table it is clear that the statement 'We should visit Neuschwanstein' is the conclusion and the picture of Neuschwanstein at dusk is the premise. As in both statements, something is attributed to the castle, we can say that Neuschwanstein is the reference of the conclusion and also of the premise. The conclusion has 'should be visited by us' as its predication and the premise has a predication depending on the interpretation of the picture.

Step 2. Rephrase the argument
This step entails a standardization of the argument in the form 'conclusion because premise', with both statements consisting of a reference and a predication. While the conclusion and premise can be taken from the KC Table, this step also

requires the analyst to break these further down into their references and predications. The standardized version of the multimodal argument provided as an example is 'Neuschwanstein should be visited by us because [Neuschwanstein] [interpretation of the predication]'. To find a suitable interpretation of the predication, the analyst can make use of insights developed in visual grammar (see Kress & van Leeuwen, 1996).

Step 3. Determine the argument form
In this step, the analyst determines which of the four argument forms reflects the configuration of references and predications in the statements expressed in the conclusion and the premise of the argument. For this purpose, the decision tree pictured in Figure 1 can be used. In this case, the argument form is alpha (a is X because a is Y), as the conclusion and the premise share the same reference, namely Neuschwanstein.

Step 4. Determine the argument substance
Given that the argument is of the alpha form, the analyst in this step must determine the types of predication expressed in the conclusion and the premise of the argument. To do so, as explained in the previous section, definitions and examples of the three predication types (fact, value, and policy) are available. In the case of the example, the conclusion is a statement of policy (P) and, without taking the context in which the argument is put forward into account, the premise can express any type of statement. But this does not mean the analyst can't try to justify their interpretation. If the conclusion, for instance, would be something like 'Neuschwanstein has four big towers', the picture probably expresses a statement of fact. In this case, as it is relevant for visiting a castle that it is worthwhile visiting, the picture probably expresses a statement of value (aesthetic statements being a subtype of statements of value). Note that for identifying the type of argument it is not necessary to verbalize the statement (e.g. by reconstructing it as 'Neuschwanstein is beautiful' or 'Neuschwanstein is impressive') but only to determine the type of predication.

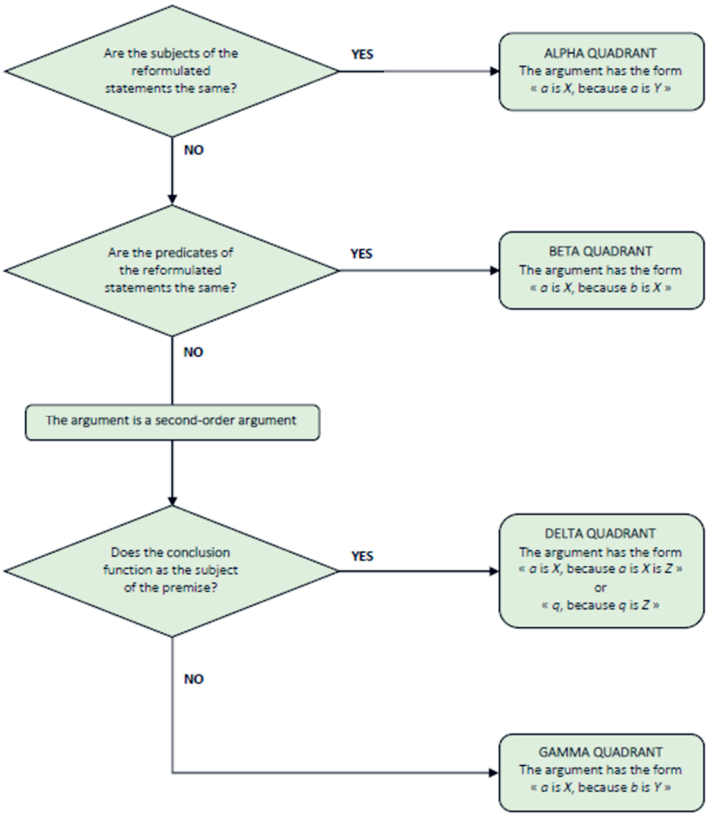


Fig. 1: Decision tree for determining the argument form (Wagemans, 2021, p. 8; 2023, p. 124)

Step 5. Determine the argument lever

The argument lever is defined as the relationship between the non-common elements of the conclusion and the premise. Its concrete expression can be derived from the results of the previous steps. In the case of the Neuschwanstein argument, which has an alpha form, the lever is the relationship between the predication of the premise and that of the conclusion. Based on the argument substance, which is PV, the analyst can formulate the lever by connecting the two predications with a keyword expressing the underlying mechanism of the argument. This would yield '[interpretation of the predication] is an evaluative reason for being visited by us'.

Step 6. Name the argument

In this step, the analyst summarizes the characteristics determined in the previous steps by labeling the argument with a name. In this case, the 'argument from evaluation' seems a good candidate. In Figure 2, I have pictured the PTA and encircled the relevant argument type.

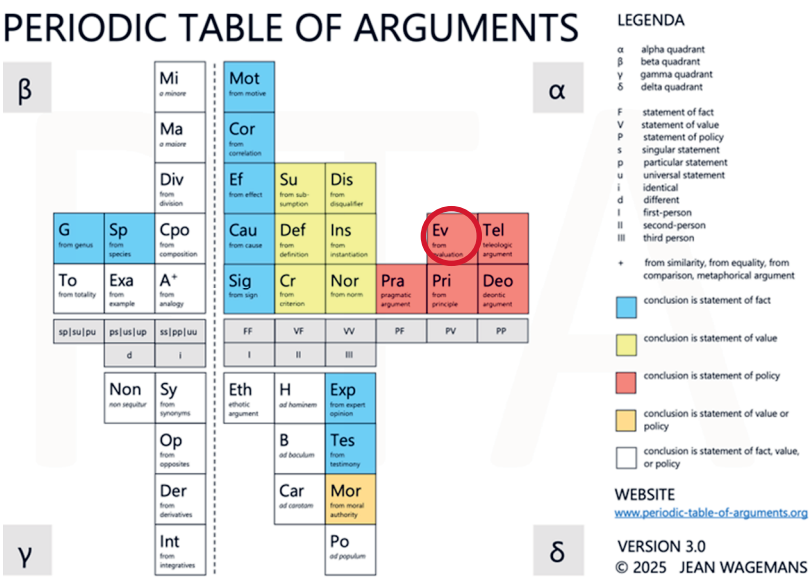


Fig. 2: The Periodic Table of Arguments with the argument from evaluation encircled in red.

As indicated in Section 1, the main reason for identifying the type of a given argument is to be able to specify the 'points of attack' functioning as the objects of evaluation of the quality of the argument under scrutiny. The output of the identification procedure can thus be used as an input for an evaluation procedure (see also Hinton & Wagemans, 2022).

Following the informal-logic tradition, arguments can be evaluated by asking so-called 'critical questions' pertaining to the acceptability of the premise content, the relevance of that premise, which is expressed in the lever, and the sufficiency of the premise in supporting the conclusion (Hornikx & Wagemans, 2023, pp. 186–188). Below, I will briefly indicate which questions can be generated from the identification of the Neuschwanstein argument as an argument from evaluation.

1. Acceptability question
   Is Neuschwanstein really [interpretation of the predication]?
2. Relevance question
   Is the fact that Neuschwanstein is [interpretation of the predication] an evaluative reason for being visited by us?
3. Sufficiency question
   Is the fact that Neuschwanstein is [interpretation of the predication] enough reason for being visited by us?

As illustrated above, the first question pertains to the acceptability of the premise. Its content is the point of attack, and the question can be generated by putting a question mark behind the premise. The second question pertains to the relevance of the premise, which is expressed by the lever. This question can thus be generated by putting a question mark behind the lever. The third and last question is usually only asked in case the acceptability and relevance questions are answered in the positive. For even if the premise content is acceptable and the premise provides relevant support for the conclusion, this support might not suffice to accept the conclusion.

Without explicating these three questions in a systematic way, Groarke in his suggestions for evaluating the Neuschwanstein argument comes up with concrete objections pertaining to exactly the same points of attack:

> Like verbal premises, a visual premise may be rejected. In the current case, someone might claim the photograph was 'doctored' in some way, taken on a rare evening, or from a vantage point that makes it a poor indication of what one is likely to see when one visits Neuschwanstein. Even if one accepts the photograph as a reliable account of what is likely to see there, one might reject the proposed conclusion by arguing that it doesn't follow (because there are other sites to visit that are, for example, more spectacular or more historically interesting). (2019, pp. 347–348).

While the objection that the photograph was doctored pertains to the acceptability of the premise content, the claims that it was taken on a rare evening or from a vantage point that makes it a poor indication of what one is likely to see both pertain to the relevance of the premise. Finally, the counterargument that there are other, more spectacular or interesting sites to visit, pertains to its sufficiency.

5 Conclusion

In this chapter I have introduced the Multimodal Argument Type Identification Procedure (MATIP), a method for identifying argument types that incorporate non-verbal components based on the argument categorization framework of the Periodic Table of Arguments (PTA). After having indicated my position in the general academic debate about various aspects of multimodal argumentation, I have discussed existing taxonomies that are claimed to be suitable for the purpose of identifying multimodal argument types. I argued they suffer from several internal problems as well as from two more general problems that I have named the 'matching problem' and the 'equivalence problem'. Consequences of these problems are that the identification of argument types may vary considerably in quality, may be incomplete or imprecise, and runs the risk of being too subjective, resulting in low inter-annotator agreement.

The PTA framework and its associated MATIP avoid these consequences for several reasons, which can be explained at both the theoretical and methodological levels. Regarding the theoretical starting points, the PTA employs a concept of 'statement' that allows for crossmodal aggregation of reference-predication pairs. This means the characteristics of a given argument can be determined independently of the mode in which the individual elements constituting that argument are expressed – whether verbal, visual, or otherwise. At the methodological level, the procedure used for identifying the type of argument introduces significant advantages. Unlike other methods, it eliminates the necessity of translating non-verbal information into equivalent verbal expressions. This does not only avoid potential inaccuracies introduced during translation (the equivalence problem) but also reduces the inherent difficulties associated with reconstructing arguments through alignment with predefined argument schemes (the matching problem). Crucially, the procedure allows the type of argument to be identified in a way that preserves the full semantic richness of non-verbal elements. This preservation is particularly important during the subsequent evaluation stage. Here, the method enables considering the emotional and visceral impact of non-verbal information – such as the evocative power of images – when addressing critical questions about the argument.

For conducting empirical research into the use of specific argument types or any particular characteristics they exhibit, the availability of larger corpora of annotated arguments is a prerequisite. While the PTA has proven to be effective for annotating verbal arguments (see, e.g., Visser et al., 2021), the development of the MATIP extends this capability to the realm of multimodal arguments. Because the PTA employs a parametric approach to argument categorization, their charac-

teristics can be identified independently, allowing for greater flexibility in research design and enhancing the precision of descriptive research.

Additionally, the application of the MATIP to create larger corpora of multimodal argumentation can serve as an initial step for normative research endeavors, where the critical evaluation of the argumentation is a central objective. The assessment procedure proposed in this paper only takes the aspect of argumentation called 'reasoning' into account by asking critical questions pertaining to the acceptability, relevance, and sufficiency of an individual argument. The procedure can be extended by including questions pertaining to other aspects of argumentation, such as 'expression' and 'process' (see Hinton & Wagemans, 2022, p. 80).

The PTA focuses on analyzing arguments at the level of individual units, which can be considered as distinct argumentative 'atoms'. To enable an analysis of larger argumentative discourse, the PTA framework can be integrated with insights from research into complex argumentation structures. These structures, which can be thought of as 'argumentative molecules', represent specific and recurring configurations or concatenations of individual arguments. By studying these configurations, researchers can identify patterns that reveal preferred or frequently occurring ways in which arguments are organized to achieve particular communicative goals or effects, thereby developing a more comprehensive model of argumentation that bridges the micro-level analysis of individual arguments with the macro-level understanding of the overall structure and content of the discourse.

Acknowledgement: I would like to thank the attendants of the ECA 2019 Conference Panel "Multimodal Dissent: Argument Schemes and Critical Questions" at the University of Groningen, June 27, 2019, the workshop "Multimodal Argumentation: Text, Image, Sound, and Gesture in Persuasive Communication" at the University of Salzburg, April 20, 2023, and the workshop "Multimodal Argumentation: Bridging Linguistics, Semiotics & Philosophy" at the University of Groningen, June 13–14, 2024, for their helpful comments on earlier presentations of my thoughts on identifying multimodal argument types. I am also grateful to Hartmut Stöckl and an anonymous reviewer for their constructive criticisms on an earlier version of this chapter.

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